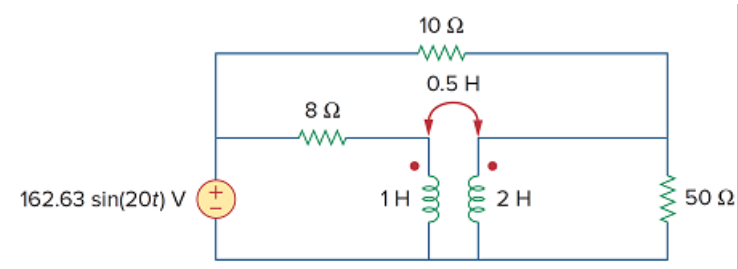


Problem

Find the average power delivered to the $50\text{-}\Omega$ resistor in the circuit of Fig. 13.96.

Figure 13.96



Step-by-step solution

Step 1 of 6

Refer to the circuit in Figure 13.96 in the textbook.

Convert the circuit into its frequency-domain equivalent circuit.

$$1\text{ H} \Rightarrow j\omega L_1 = j20\ \Omega$$

$$2\text{ H} \Rightarrow j\omega L_2 = j40\ \Omega$$

$$0.5\text{ H} \Rightarrow j\omega M = j10\ \Omega$$

The frequency-domain equivalent of the circuit is,

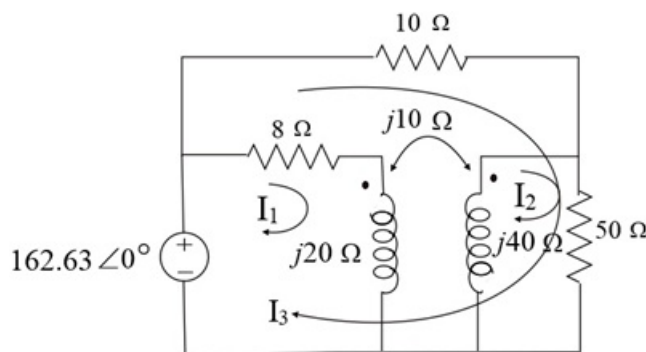


Figure 1

[Comments \(1\)](#)



Anonymous

why divide by root 2?

Step 2 of 6

Apply Kirchhoff's voltage law in mesh 1.

$$(8 + j20)I_1 - j10I_2 = \frac{162.63}{\sqrt{2}}$$

$$(8 + j20)I_1 - j10I_2 = 115 \dots\dots (1)$$

Apply Kirchhoff's voltage law in mesh 2.

$$-j10I_1 + j40I_2 + 50(I_2 + I_3) = 0$$

$$-j10I_1 + (50 + j40)I_2 + 50I_3 = 0 \dots\dots (2)$$

Apply Kirchhoff's voltage law in mesh 3.

$$50I_2 + 60I_3 = 115 \dots\dots (3)$$

Step 3 of 6

Put equations (1), (2), and (3) in matrix form.

$$\begin{bmatrix} 8 + j20 & -j10 & 0 \\ -j10 & 50 + j40 & 50 \\ 0 & 50 & 60 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 115 \\ 0 \\ 115 \end{bmatrix}$$

Find the value of the determinant.

$$\begin{aligned} \Delta &= \begin{vmatrix} (8 + j20) & -j10 & 0 \\ -j10 & (50 + j40) & 50 \\ 0 & 50 & 60 \end{vmatrix} \\ &= (8 + j20)[(50 + j40)60 - (50 \times 50)] + j10(-j10 \times 60 - 50 \times 0) \\ &= (8 + j20)[500 + j2400] + 6000 \\ &= 4000 - 48000 + j10000 + j19200 + 6000 \\ &= -38000 + j29200 \\ &= 47923.27 \angle 142.46^\circ \end{aligned}$$

Step 4 of 6

Find the value of Δ_2 .

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} (8 + j20) & 115 & 0 \\ -j10 & 0 & 50 \\ 0 & 115 & 60 \end{vmatrix} \\ &= (8 + j20)[0 - (115 \times 50)] - 115(-j10 \times 60 - 0) \\ &= (8 + j20)[-5750] + j69000 \\ &= -46000 - j115000 + j69000 \\ &= -46000 - j46000 \\ &= 65053.82 \angle -135^\circ \end{aligned}$$

Step 5 of 6

Determine the current I_2 .

$$\begin{aligned} I_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{65053.82 \angle -135^\circ}{47923.27 \angle 142.46^\circ} \\ &= 1.35 \angle 82.54^\circ \text{ A} \\ &= 0.175 + j1.33 \text{ A} \end{aligned}$$

Step 6 of 6

Find the value of Δ_3 .

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} (8 + j20) & -j10 & 115 \\ -j10 & 50 + j40 & 0 \\ 0 & 50 & 115 \end{vmatrix} \\ &= (8 + j20)[(115)(50 + j40)] + j10(-j10 \times 115 - 115 \times 50) \end{aligned}$$

Calculate I_3 .

$$\begin{aligned} I_3 &= \frac{\Delta_3}{\Delta} \\ &= \frac{(8 + j20)[(115)(50 + j40)] + j10(-j10 \times 115 - 115 \times 50)}{47923.27 \angle 142.46^\circ} \\ &= 1.768 - j1.1215 \text{ A} \end{aligned}$$

Calculate the power delivered to the 50 Ω resistor.

$$\begin{aligned} P &= (I_2 + I_3)^2 \times 50 \\ &= (1.946 + 0.17)^2 \times 50 \\ &= 191.86 \text{ W} \end{aligned}$$

Thus, the power delivered to the 50- Ω resistor is 191.86 W.